

CMPS 460 Machine Learning Project – Spring 2026

Due by 11:59pm on Sunday, 3th May 2026

The goal of this project is to apply the complete machine learning lifecycle to a real-world dataset, including data exploration, preprocessing, model training, evaluation, and optimization.

Deliverable: One well-organized **Jupyter Notebook** per team that includes:

0. Team Formation

You should work on this project in teams of 3 persons each. If you form a team, you need all the team members should self-enroll themselves in one (the same) empty Blackboard group (available under “المجموعات”).

1. Dataset Selection

Work on the dataset assigned to your group. The dataset is a Gene Expression Matrix where rows are the samples (the patients or tissue biopsies) and the columns are the genes/features (the RNA-seq reads).

What the Components Represent

- **Rows (The Samples; dataset_EX):** Each row represents a single biological observation. This could be a tissue biopsy from a specific patient, a blood draw, or a cell line culture.
- **Columns (The Features):** Each column represents a specific gene (humans have roughly 20,000 protein-coding genes).
- **The Values (RNA-seq Reads):** The numbers inside the cells represent the **gene expression level**. During RNA sequencing, RNA molecules are captured, converted to cDNA, and sequenced into "reads." If a cell has a high number of reads mapped to a specific gene (like *EGFR* in Patient 1), it means that gene is highly active, or "upregulated," in that sample.
- **The Target/Label Column (dataset_Label):** This is your metadata. It defines the "class" of the sample—in this case, binary classification: Cancerous (1: Tumor) or Non-Cancerous (0: Healthy/Normal control).

2. Exploratory Data Analysis (EDA)

- Describe dataset structure and features.
- Use summary statistics and meaningful visualizations, then draw meaningful observations.
- Identify distributions, correlations, patterns, or anomalies.

3. Data Cleaning and Preprocessing

- Handle missing values, outliers, and duplicates.
- Perform feature engineering guided by correlation analysis.
- Split data into training and testing sets.
- For a large dataset, you may use stratified sampling when appropriate.

4. Model Design and Training

Train **at least three models**, each from a different category:

- **Traditional ML Model** (e.g., Linear Regression, Decision Tree, Naïve Bayes)
- **Ensemble Model** (e.g., Random Forest, XGBoost)
- **Deep Learning Model** (e.g., Neural Network)

For each model:

- Justify why it is appropriate for the dataset.
- Describe architecture and key parameters used.

5. Model Evaluation and Comparison

- Evaluate models using appropriate metrics (e.g., accuracy, F1-score, MSE).
- Compare performance across models.

- Analyze strengths, weaknesses, and error patterns.

6. Model Optimization

- Apply optimization techniques such as:
 - Hyperparameter tuning
 - Regularization
 - Feature selection or ranking
- Demonstrate and discuss performance improvements.

7. Documentation and Reporting

Your Jupyter Notebook must include:

- Clear section headings and logical flow
- Well-commented code and explanatory Markdown cells
- Justification for all major decisions
- Interpretation of results and final conclusions

Oral Discussion

Each team **must** attend a minimum of **10-minute discussion** during office hours to present the project and answer questions to confirm understanding and authorship. This implies that different members of the same team may get different grades.

Project Grading Rubrics

Requirement	Poor (0-50%)	Fair (51-65%)	Good (66-85%)	Excellent (86-100%)
Exploratory Data Analysis (EDA) (5%)	- Minimal EDA; few or no meaningful statistics or visualizations. - Little understanding of the dataset's structure, distributions, and relationships. - Minimal or absent interpretation of findings; lacks depth and thoroughness.	- Basic EDA with some summary statistics and visualizations. - Partial understanding of the dataset's structure, distributions, and relationships. - Limited or basic interpretations of findings.	- Good EDA using appropriate summary statistics and visualizations. - Clear understanding of the dataset's structure, distributions, and relationships. - Reasonable interpretations of findings including patterns or anomalies based on the analysis.	- Comprehensive EDA with well-chosen summary statistics and clear visualizations. - Deep insight into the dataset's structure, distributions, and relationships. - Strong interpretation of findings including patterns or anomalies based on the analysis.
Data Cleaning and Preprocessing (10%)	- Little to no addressing of missing values, outliers, and duplicates in the dataset. - Limited or simplistic feature engineering.	- Basic handling of missing values, outliers, and duplicates. - Weak feature engineering or lack depth and justifications.	- Effectively handling of missing values, outliers, and duplicates using mostly appropriate techniques. - Reasonable feature engineering with justifications.	- Robust and well-justified preprocessing using appropriate techniques. - Thorough feature engineering with clear justifications.
Model Design and Training (25%)	- Models poorly implemented or incomplete; missing required model categories. - Little or no justification. - Little or no explanation of architectures and parameters.	- Two or three models trained but lacking depth, correctness, or justification; weak alignment with dataset. - Weak explanation of architectures and parameters.	- Correct implementation of all three model types; clear and reasonable justification based on dataset and task. - Reasonable explanation of architectures and parameters.	- Expertly designed and implemented models; insightful justification. - Well-explained architectures and parameters.

Model Evaluation and Comparison (10%)	- Inadequate or incorrect evaluation. - Little or no comparison between models. - Limited or no analysis of strengths and weaknesses of each model.	- Basic evaluation using limited metrics. - Basic comparison and analysis of results. - Basic analysis of strengths and weaknesses of each model.	- Complete model evaluation using appropriate metrics. - Clear comparison and meaningful analysis of results. - Detailed analysis of strengths and weaknesses of each model.	- Comprehensive evaluation with multiple metrics. - Deep analysis of performance, errors, and trade-offs across models. - In-depth analysis of strengths and weaknesses of each model.
Model Optimization (15%)	Little or no optimization attempted; no measurable improvement.	Basic optimization applied; moderate improvement with limited explanation.	Appropriate optimization techniques applied (e.g., tuning, regularization); clear performance gains.	Multiple optimization strategies applied effectively; superior performance with strong technical justification.
Notebook Quality & Organization (5%)	Disorganized notebook; unclear flow; poorly commented or unreadable code.	Basic structure present; some comments and explanations but lacking clarity or consistency.	Well-organized notebook; clear sections; readable, well-commented code and explanations.	Exceptionally organized and professional; excellent clarity, documentation, and presentation throughout.
Project Discussion (30%)	Has a limited or inaccurate project understanding. Cannot clearly explain their own or others' contributions. Provides unclear or inaccurate answers, even about their own responsibilities.	Shows basic project understanding. Describes own work but has limited awareness of others' roles. Answers questions about their own work but struggles with questions on other project components.	Has a solid understanding of the project. Describes own contributions and is aware of others' roles. Answers most questions accurately, but knowledge of other components may be more general.	Shows deep understanding of the entire project. Clearly explains own and others' contributions. Confidently and insightfully answers questions about all components, even those they did not work on.